

Mark schemes

Q1.

Marking Instructions	AO	Marks	Typical Solution
Selects correct answer	AO1.1b	B1	$(-1, 8)$
Total 1 mark			

Q2.

Marking Instructions	AO	Marks	Typical Solution
Selects correct answer	AO1.1b	B1	$\frac{2}{3}$
Total 1 mark			

Q3.

Marking Instructions	AO	Marks	Typical Solution
Forms an equation for gradient of $CD = \frac{1}{4}$ or $-\frac{1}{4}$ of the form difference in y over difference in x (or vice versa = 4 or -4)	AO3.1a	M1	$\frac{d-2}{6-c} = \frac{1}{4}$ $4d - 8 = 6 - c$ $c + 4d = 14$ $\frac{2+d}{2} + 4\left(\frac{c+6}{2}\right) = 11$ $4c + d = -4$ $c = -2 \qquad d = 4$
Obtains a correct equation for c & d	AO1.1b	A1	
Forms an equation for the mid-point of CD lying on y + $4x = 11$	AO3.1a	M1	
Obtains correct equation for c & d (any correct form)	AO1.1b	A1	
Solves for c and d CAO	AO1.1b	A1	
			Total 5 marks

Q4.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Selects a method leading to any calculation pertaining to one of the following methods seen (not necessarily correct); gradients of sides, lengths of sides or intersection or lengths of diagonals	AO3.1a	M1	Grad $BC = -5/2 = \text{Grad } DA$ Grad $AB = 2/5 = \text{Grad } DC$ Both pairs of opposite sides have equal gradient so parallel, so $ABCD$ is a parallelogram
	Finds gradients of all 4 sides or lengths of all 4 sides or midpoints of both diagonals correctly	AO1.1b	A1	Grad $BC \times \text{grad } AB = -1$ $ABC = 90^\circ$ therefore all angles in $ABCD$ are 90° so $ABCD$ is a rectangle
	Proves one angle is 90° by using gradients or Pythagoras	AO1.1a	M1	
	Completes proof that $ABCD$ is a rectangle. There must be a clear statement that there are 2 pairs of parallel sides and all angles are 90°	AO2.1	R1	
Note, there are various ways of proving that $ABCD$ is a rectangle (1 – 5 below score M1 A1 M1 before final required statement for relevant R1 stating how their method used proves a rectangle) - As in the typical solution shown: show that both pairs of opposite sides are parallel, show that one angle is 90°. - Show that each pair of opposite sides is equal in length, show that one angle is 90°. - Show that one pair of opposite sides is parallel and equal in length, show that one angle is 90°. - Show that the diagonals bisect (the midpoint of one is also the midpoint of the other) and are equal in length. - Show that each pair of opposite sides are parallel and length of the two diagonals are the same NB May be expressed using vectors NB Diagonals AC and $BD = \sqrt{377}$				
(b)	Finds correct lengths of two adjacent sides (accept to at least 1dp accuracy)	AO1.1a	M1	$AB (= DC) = \sqrt{261} = 3\sqrt{29}$ $BC (= DA) = \sqrt{116} = 2\sqrt{29}$
	Obtains correct area (AWRT)	AO1.1b	A1	Area = 174
				Total 6 marks

Q5.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Uses a technique which could lead to showing two lines are perpendicular. Obtains at least one correct distance (or distance ²) or gradient.	AO3.1a	M1	$AB^2 = (8 - 15)^2 + (17 - 10)^2$ $= 98$ $AC^2 = (8 - -2)^2 + (17 - -7)^2$ $= 676$ $CB^2 = (15 - -2)^2 + (10 - -7)^2$ $= 578$ $AB^2 + BC^2 = 98 + 578$ $= 676$ $= AC^2$ Angle ABC is a right angle.
	Obtains three correct distances (or distance ²) or two gradients. Lengths: $7\sqrt{2}, 17\sqrt{2}, 26$ Gradients: $AB = -\frac{7}{7}, BC = \frac{17}{17}$	AO1.1b	A1	
	Completes correct rigorous argument to show required result Uses Pythagoras OR Multiplies gradients to show product is -1 AND Writes a concluding statement.	AO2.1	R1	
(b)	Explains why AC is a diameter	AO2.4	E1	The angle subtended by a diameter is 90° ∴ AC must be a diameter of the circle
(i)	Must reference angle subtended by diameter (condone "angle in a semi-circle") or give full explanation.			
	Deduces correct radius (or radius ²)	AO2.2a	B1	Radius $\frac{\sqrt{676}}{2} = 13$
	Obtains mid-point of diameter	AO1.1b	B1	Centre $\left(\frac{8-2}{2}, \frac{17-7}{2}\right) = (3, 5)$
	Uses D(-8, -2) to find the distance or (distance ²) from their centre OE	AO1.1a	M1	Distance from centre to D $(3 - -8)^2 + (5 - -2)^2 = 11^2 + 7^2$ $= 170$
(ii)	Completes rigorous argument by comparing $\sqrt{170} > 13$ (or $170 > 169$) to show that D lies outside the circle	AO2.1	R1	> 169 So D lies outside the circle.
				Total 8 marks

Q6.

Marking Instructions	AO	Marks	Typical Solution
Circles correct answer	AO1.1b	B1	$3x - 2y = 7$
Total 1 mark			

Q7.

Marking Instructions	AO	Marks	Typical Solution
Explains that equal gradients implies that lines are parallel	AO2.4	E1	Parallel lines have equal gradient
Finds the gradient of the given line CAO	AO1.1b	B1	$2x + 3y + 4 = 0 \Rightarrow y = -\frac{2}{3}x - \frac{4}{3}$ So gradient is $-\frac{2}{3}$
Finds the gradient of the line through the 2 given points CAO	AO1.1b	B1	Gradient of line through (9, 4) and (3, 8) is $\frac{8-4}{3-9} = -\frac{2}{3}$
Deduces that the two lines are parallel	AO2.2a	R1	So line with equation $2x + 3y + 4 = 0$ is parallel to the line joining the points with coordinates (9, 4) and (3, 8) as both have gradient $-\frac{2}{3}$
Total 4 marks			

Q8.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Obtains (at least four) correct $\log_{10}y$ values, in table or plotted	AO1.1a	M1	(1, 1.1) (2, 1.7) (3, 2.1) (4, 3.0) (5, 3.1) (6, 3.5)
	Plots all points correctly	AO1.1b	A1	(Points above plotted on grid)

(b)	Identifies $y = 1100$ and gives correct reason	AO2.2b	B1	(4, 1100), as it is not on the line that the other points are close to
(c)	Uses laws of logs. (May earn in part (a) if laws of logs were used there)	AO1.1a	M1	$\log_{10} y = \log_{10} k + x \log_{10} b$ Vertical intercept $c = 0.68$ $(= \log_{10} k)$ Therefore from intercept: $k = 10^{0.68}$
	Draws straight line and calculates/measures the vertical intercept c and attempts 10^c or calculates/measures gradient m and attempts 10^m Alternatively uses regression line from calculator to get intercept and gradient	AO1.1a	M1	Gradient $m = 0.48 = \log_{10} b$ Therefore from gradient: $b = 10^{0.48}$
	Finds correct value of b from 'their' gradient, provided $0.45 < \text{'their' gradient} < 0.51$	AO1.1b	A1F	$k = 4.8$
	Finds correct value of k from 'their' intercept, provided $0.6 \leq \text{'their' intercept} \leq 0.8$	AO1.1b	A1F	$b = 3.0$
				Total 7 marks

Q9.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Rewrites given expression with a fractional power and negative power – at least one index form must be correct	AO1.1a	M1	$y = 6x^{\frac{3}{2}} + 32x^{-1}$
	Both terms correct	AO1.1b	A1	

(b)	Differentiates 'their' rewritten expression – at least one term correct	AO1.1a	M1	$\frac{dy}{dx} = 6 \times \frac{3}{2} \times x^{\frac{1}{2}} - 32x^{-2}$ $= 9\sqrt{x} - \frac{32}{x^2}$
	Both terms correct for 'their' expression	AO1.1b	A1F	
	Finds the equation of the tangent, a clear attempt must be seen	AO3.1a	M1	When $x = 4$,
	Evaluates 'their' $\frac{dy}{dx}$ (from part (a)) correctly (when $x = 4$)	AO1.1b	A1F	$\frac{dy}{dx} = 9 \times 2 - \frac{32}{16} = 16$ and
	Obtains correct y value (when $x = 4$)	AO1.1b	A1	$y = 6 \times 4 \times 2 + \frac{32}{4} = 56$
	Obtains correct form of the equation of a straight line using 'their' values for y and $\frac{dy}{dx}$	AO1.1b	A1F	Tangent: $y - 56 = 16(x - 4)$ When $y = 0$, $x = 4 - \frac{56}{16} = 0.5$ (0.5,0)
	Deduces value required at x -axis is when y equals 0 (follow through from 'their' equation) Both coordinates needed, any form	AO2.2a	A1F	
				Total 9 marks

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Q10.

	Marking Instructions	AO	Marks	Typical Solution
(a)	States correct radius CAO	AO1.2	B1	Radius = $\sqrt{5}$
(i)				
(ii)	States correct centre CAO	AO1.2	B1	C is (7, -2)
(b)	Finds gradient of the line through the points P and 'their' C (as found in part (a)) Condone one sign error	AO3.1a	M1	Gradient $CP =$ $CP = \frac{-1 - (-2)}{5 - 7} = -\frac{1}{2}$

(c)	Correct tangent gradient obtained from 'their' CP gradient	AO3.1a	M1	So tangent gradient = 2
	Uses a correct form for the equation of a straight line with correct coordinates of P and 'their' tangent gradient	AO1.1a	M1	$y - (-1) = 2(x - 5)$
	States correct final answer in required form ($y = mx + c$) FT from 'their' C found in part (a)	AO1.1b	A1F	$y = 2x - 11$
	Identifies QTC as a right-angled triangle PI	AO3.1a	M1	QTC is a right-angled triangle so we can use Pythagoras
	Finds QC or QC^2 FT 'their' C found in part (a)	AO1.1b	B1F	$QC^2 = (7 - 3)^2 + (-2 - 3)^2$
	Uses Pythagoras' theorem correctly for 'their' triangle	AO1.1a	M1	$4^2 + 5^2 = (\sqrt{5})^2 + QT^2$
	Correct evaluation of length of QT FT 'their' QC and 'their' radius found in part (a)	AO1.1b	A1F	$QT^2 = 36$ so $QT = 6$
				Total 10 marks

Q11.

Marking Instructions	AO	Marks	Typical Solution
Circles correct answer	AO1.1b	B1	$-\frac{2}{5}$
			Total 1 mark

Q12.

Marking Instructions	AO	Marks	Typical Solution
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(a)	States correct expression for a	AO1.1b	B1	$a = \frac{V-U}{T}$
(b)	Rearranges to make T the subject of the formula	AO2.1	R1	$T = \frac{V-U}{a}$
	Uses given expression for S and attempts to eliminate T	AO2.1	R1	$S = \frac{1}{2}(U+V) \times \frac{V-U}{a}$ $2as = (U+V)(V-U)$
	Completes argument to reach required result AG Only award if they have a completely correct solution, which is clear, easy to follow and contains no slips	AO2.1	R1	$V^2 = U^2 + 2aS$ (AG)
Total 4 marks				

Q13.

(a) (i) $21 + 5k = 1$
condone $3 \times 7 + 5k = 1$
 $\Rightarrow k = -4$
AG *condone* $y = -4$

B1

1

(ii) $(x =) 2$

B1

$(y =) -1$

midpoint coords are $(2, -1)$

B1

2

(b) $y = \frac{1}{5} - \frac{3}{5}x$

obtaining $y = a \pm \frac{3}{5}x$ or $\frac{\Delta y}{\Delta x} = \frac{-4-2}{7-3}$ or $\frac{-1-2}{2-3}$ or $\frac{-4-1}{7-2}$
condone one sign error in expression

M1

(Gradient $AB =) -\frac{3}{5}$

allow -0.6 , $\frac{6}{-10}$ etc for A1 & condone error in rearranging if gradient is correct.

A1

2

- (c) Perp grad = $\frac{5}{3}$
 $-1 / \text{"their" grad } AB$

M1

$$\left. \begin{array}{l} y - 2 = \frac{5}{3}(x + 3) \\ \text{or } y = \frac{5}{3}x + c, \quad c = 7 \quad \text{etc} \end{array} \right\}$$

correct equation in any form (must simplify $x - -3$ to $x + 3$ or c to a single term equivalent to 7)

A1

$$5x - 3y + 21 = 0$$

or any multiple of this with integer coefficients – terms in any order but all terms on one side of equation

A1

3

- (d) $3x + 5y = 1$ and $5x + 8y = 4 \Rightarrow Px = Q$ or $Ry = S$
 must use **correct** pair of equations **and** attempt to eliminate y (or x) (generous)

M1

$$x = 12$$

A1

$$y = -7$$

$$(12, -7)$$

A1

3

[11]

Q14.

- (a) (i) $2x - 2y \frac{dy}{dx} = 0$
 Correct differentiation

M1

$$\frac{dy}{dx} = \frac{x}{y}$$

$$\text{at } (p, q) \quad \frac{dy}{dx} = \frac{p}{q}$$

(p, q) substituted into correct derivative or $x = p$ $y = q$ stated

AG

A1

2

Alternative:

$$y = \sqrt{x^2 - 8} \quad \frac{dy}{dx} = \frac{1}{2} \times 2x(x^2 - 8)^{-\frac{1}{2}} = \frac{x}{y}$$

(M1)

$$= \frac{p}{q}$$

(A1)

(2)

Alternative:

$$\frac{dy}{dt} = 1 + \frac{2}{t^2} \quad \frac{dx}{dt} = 1 - \frac{2}{t^2}$$

Attempt parametric derivatives and use chain rule.

(M1)

$$\frac{dy}{dx} = \frac{1 + \frac{2}{t^2}}{1 - \frac{2}{t^2}} = \frac{t + \frac{2}{t}}{t - \frac{2}{t}} = \frac{x}{y}$$

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$$\text{at } (p, q) \quad \frac{dy}{dx} = \frac{p}{q}$$

(p, q) substituted into correct derivative.

(A1)

(2)

(ii) tangent at (p, q) $y - q = \frac{p}{q}(x - p)$
ACF

B1

tangent at $(p, -q)$ $y - (-q) = \frac{-p}{q} (x - p)$
 ACF

B1

add $2y = 0$
 Solve tangent equations for y .

M1

conclusion $y = 0 \Rightarrow$ intersect on Ox
 Conclusion required

A1

4

Alternative:

tangent at (p, q) $y - q = \frac{p(x - p)}{q}$
 ACF

(B1)

tangent at $(p, -q)$ $y - (-q) = \frac{-p(x - p)}{q}$
 ACF

(B1)

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When $y = 0$ $\frac{-q^2}{p} = x - p$ and $\frac{q^2}{-p} = x - p$
 Substitute $y = 0$ into both candidate's tangents
 and solve for x .

(M1)

$x = p - \frac{q^2}{p}$ is on both lines, so intersect on x axis
 Conclusion

(A1)

$x + y = 2t$ $x - y = \frac{4}{t}$

(4)

(b) $x^2 = t^2 + 4 + \frac{4}{t^2}$ $y^2 = t^2 - 4 + \frac{4}{t^2}$

Attempt to square x and y and subtract.

M1

$$x^2 - y^2 = 8$$

All correct AG Allow $8 = 8$

A1

2

Alternative:

$$(x - y)(x + y) = 2t \times \frac{4}{t}$$

Attempt to eliminate t

(M1)

$$x^2 - y^2 = 8$$

(A1)

(2)

[8]

Q15.

(a) $3p - 4(p + 2) + 5 = 0$

condone omission of brackets or one sign error

M1

$$(\Rightarrow p =) -3$$

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A1

2

(b) $y = \frac{3}{4}x + \frac{5}{4}$

rearranging into form $y = \pm \frac{3}{4}x + c$

M1

$$(\text{gradient } AB =) \frac{3}{4}$$

condone slips in rearranging if gradient is correct.

A1

2

(c) (gradient $AC = \frac{k-2}{-5-1}$
 or $\frac{2-k}{1-5}$ (condone one sign error)

M1

“their” $\frac{(k-2)}{-6} \times \frac{3}{4} = -1$ OE
 product of grads = -1 in terms of k

m1

$(\Rightarrow k =) 10$

A1

3

(d) $3x - 4y + 5 = 0$ and $2x - 5y = 6$
 $\Rightarrow P \ x = Q$ or $R \ y = S$
 must use “correct” pair of equations **and**
 attempt to eliminate y (or x) (generous)

M1

$x = -7$

A1

$y = -4$

A1

$(-7, -4)$

3

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[10]

Q16.

(a) $\frac{dy}{dx} = 5x^4 - 4x$
 one of these terms correct

M1

all correct (no + c etc)

A1

$(= 5(-1)^4 - 4(-1)) = 9$

A1

Tangent has equation $y = \text{'their' } 9x + c$

and $6 = \text{'their' } 9(-1) + c \Rightarrow = \dots$

*tangent using 'their' gradient, and attempt to find c
using $x = -1$ and $y = 6$*

m1

$$\Rightarrow y = 9x + 15$$

equation must be seen in this form

A1

5

- (b) (i) When $x = 2$, $y = 2^5 - 2 \times 2^2 + 9 = 32 - 8 + 9 = 33$

*be convinced that they are using **curve** equation*

$$(k =) 33$$

NMS $k = 33$ scores B0

B1

1

- (ii) When $x = 2$, $y = 9 \times 2 + 15 = 33$ so lies on tangent

*be convinced that they are using **tangent** equation
and have statement*

B1

1

(c) (i) $\frac{x^6}{6} - \frac{2x^3}{3} + 9x$

one of these terms correct

M1

another term correct

A1

all correct (may have +c)

A1

$$\left[\frac{2^6}{6} - \frac{2 \times 2^3}{3} + 9 \times 2 \right] - \left[\frac{(-1)^6}{6} - \frac{2 \times (-1)^3}{3} + 9 \times (-1) \right]$$

$$\left[\frac{64}{6} - \frac{16}{3} + 18 \right] - \left[\frac{1}{6} - \frac{2}{3} - 9 \right]$$

$$= 31.5$$

$F(2) - F(-1)$ unsimplified FT "their terms" from

$$\text{integration} = \frac{70}{3} - \left(-\frac{49}{6} \right)$$

m1

$$\left(\text{or } \frac{189}{6} \text{ etc} \right)$$

condone single fraction

A1

5

(ii) Area of trapezium = $\frac{1}{2} \times 3 \times (6 + \text{'their' } k)$
 $= 58.5 \text{ when } k = 33$

B1 ✓

Shaded area = **Trapezium** – ‘their’ (c)(i) value

M1

$$= 27$$

OE eg $\frac{162}{6}$

A1

3

[15]

Q17.

(a) $(OA^2 =) 6^2 + (-4)^2$; $(OB^2 =) (-2)^2 + 7^2$
either correct PI by 52 or 53 seen

M1

$(OA^2 =) 52$ **and** $(OB^2 =) 53$

or $(OA =) \sqrt{52}$ **and** $(OB =) \sqrt{53}$

*both correct values 52 or $\sqrt{52}$ **and** 53 or $\sqrt{53}$ seen*

A1

$OA = \sqrt{52}$ and $OB = \sqrt{53} \Rightarrow OA < OB$
 or $OA^2 = 52$ and $OB^2 = 53$

A1

*correct working + concluding statement involving
 OA and / or OB*

3

(b) (i) $\text{grad } AB = \frac{7+4}{-2-6}$

condone one sign error

M1

$$= -\frac{11}{8}$$

A1

2

(ii)

$$\left. \begin{array}{l} y - -4 = \text{'their grad } AB'(x - 6) \\ \text{or } y - 7 = \text{'their grad } AB'(x - -2) \end{array} \right\}$$

*or $y = \text{'their grad } AB'x + c$ and attempt to find c using
 $x = 6, y = -4$ or $x = -2, y = 7$*

M1

$$y + 4 = -\frac{11}{8}(x - 6) \quad \text{OE}$$

*any correct form eg $y = -\frac{11}{8}x + \frac{34}{8}$ but must
 simplify -- to +*

A1

$$\Rightarrow 11x + 8y = 34$$

condone $8y + 11x = 34$ or any multiple of these equations

A1

(c) (grad AC) = $\frac{8}{11}$ 2025 Exam Papers Practice. All rights reserved.

FT -1 / 'their grad AB '

B1 ✓

$$\frac{4}{k-6} = \text{'their } \frac{8}{11}' \quad \text{OE}$$

$$\Rightarrow 2k - 12 = 11$$

*equating gradients; LHS must be correct and RHS
 is "attempt" at perp grad to AB*

M1

$$\Rightarrow k = \frac{23}{2}$$

$$k = 11.5 \text{ OE}$$

A1cso

Alternative:

$$\text{Eqn AC : } (y + 4) = \text{'their' } \frac{8}{11} (x - 6)$$

$$(11y = 8x - 92) \text{ AND must sub } y = 0 \text{ for M1}$$

B1✓

$$\text{or } (y - 0) = \text{'their' } \frac{8}{11} (x - k)$$

$$\text{AND must sub } x = 6, y = -4 \text{ for M1}$$

B1✓

3

[11]

Q18.

(a) $(x + 7)^2 + (y - 5)^2$

one term correct ; condone $(x - -7)^2$

M1

both terms correct with squares
and plus sign between terms

A1

$$(x + 7)^2 + (y - 5)^2 = 5^2$$

condone 25 for 5^2

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A1cao

(b) (i) $C(-7, 5)$

correct or FT 'their' circle equation

B1✓

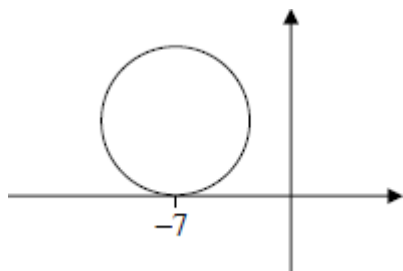
(ii) $r = 5$

correct or FT 'their' $r^2 > 0$

condone $\sqrt{25}$ etc but not $\pm \sqrt{25}$

B1✓

(c) must draw axes



freehand circle with C correct or FT 'their C' for quadrant of centre

M1

circle touching x-axis at -7 with -7 marked (need not show 5 on y-axis) but circle must not touch y-axis

A1

(d) (i) $x^2 + (kx + 6)^2 + 14x - 10(kx + 6) + 49 = 0$

clear attempt to sub $y = kx + 6$ into original or 'their' circle equation ...

$$x^2 + k^2x^2 + 12kx + 36 + 14x - 10kx - 60 + 49 = 0$$

...and attempt to multiply out

M1

$$(1 + k^2)x^2 + 2kx + 14x + 25 = 0$$

$$\Rightarrow (k^2 + 1)x^2 + 2(k + 7)x + 25 = 0$$

AG condone $x^2(1 + k^2) + 2x(7 + k) + \dots$ etc

A1cso

(ii) Equal roots ' $b^2 - 4ac = 0$ '

allow statement alone if discriminant in terms of k attempted

B1

$$[2(k + 7)]^2 - 4 \times 25(k^2 + 1)$$

discriminant (condone one slip)

M1

$$4\{k^2 + 14k + 49 - 25k^2 - 25\} = 0$$

$$-24k^2 + 14k + 24 = 0$$

$$\Rightarrow 12k^2 - 7k - 12 = 0$$

AG all working correct

but = 0 must appear before last line

A1

(iii) $(4k + 3)(3k - 4)$

correct factors or correct use of formula as far as

$$k = \frac{7 \pm \sqrt{49 + 576}}{24}$$

M1

$$\Rightarrow k = -\frac{3}{4}, k = \frac{4}{3} \quad \text{OE}$$

are values of k for which line is a tangent

A1

2

[14]

Q19.

(a) $xy^2 + 3y = (8t^2 - t)\left(\frac{3}{t}\right)^2 + 3\left(\frac{3}{t}\right)$
Substitute and expand

M1

$$= 72 - \frac{9}{t} + \frac{9}{t} = 72$$

$$k = 72$$

A1

2

Alternative

$$x = 8\left(\frac{3}{y}\right)^2 - \frac{3}{y}$$

Eliminate t

(M1)

$$xy^2 + 3y = 72$$

$$k = 72$$

(A1)

(2)

(b) (i) $\frac{dx}{dt} = 16t - 1 \quad \frac{dy}{dt} = -\frac{3}{t^2}$

B1B1

$$t = \frac{1}{4} \quad \frac{dy}{dx} = \frac{-\frac{3}{(\frac{1}{4})^2}}{16 \times \frac{1}{4} - 1}$$

Use chain rule $\left(\frac{dy}{dx} = \frac{-3}{16t^3 - t^2} \right)$ and calculate gradient using $t = \frac{1}{4}$

M1

$$= -16$$

A1

$$t = \frac{1}{4} \quad x = \frac{8}{16} - \frac{1}{4} \quad y = \frac{3}{\frac{1}{4}}$$

Calculate x and y using $t = \frac{1}{4}$

M1

$$x = \frac{1}{4} \quad y = 12$$

Both correct

A1

tangent $y = -16x + 16$
ACF CSO

$$y - 12 = -16 \left(x - \frac{1}{4} \right) \text{ ISW}$$

A1

7

Alternative

$$2xy \frac{dy}{dx} + y^2$$

Product rule attempted; two terms added, one with $\frac{dy}{dx}$

(M1A1)

$$+ 3 \frac{dy}{dx} = 0$$

(B1)

$$t = \frac{1}{4} \quad x = \frac{8}{16} - \frac{1}{4} \quad y = \frac{3}{\frac{1}{4}}$$

Calculate x and y using $t = \frac{1}{4}$

(M1)

$$x = \frac{1}{4} \quad y = 12$$

Both correct.

(A1)

$$\left(\frac{dy}{dx} = \frac{-y^2}{2xy + 3} \right) \frac{dy}{dx} = -16$$

Calculate gradient from candidate's expression.

m1

$$\text{tangent } y = -16x + 16$$

ACF CSO

$$y - 12 = -16 \left(x - \frac{1}{4} \right) \quad \text{ISW}$$

(A1)

(7)

Alternative

$$x = \frac{72 - 3y}{y^2}$$

Correct expression for x from candidate's implicit equation.

Quotient rule attempted; y^4 and two terms subtracted.

(M1)

$$\frac{dx}{dy} = \frac{y^2(-3) - (72 - 3y) \times 2y}{y^4}$$

(A1)

$$\left(\frac{dx}{dy} = \frac{3y - 144}{y^3} \right)$$

Numerator; first term; second term

(A1)

$$t = \frac{1}{4} \quad y = \frac{3}{\frac{1}{4}} = 12$$

(B1)

$$\frac{dx}{dy} = -\frac{1}{16} \quad \frac{dy}{dx} = -16$$

Use $t = \frac{1}{4}$ to calculate y

m1

$$x = \frac{1}{4} \quad x = \frac{8}{16} - \frac{1}{4} = \frac{1}{4}$$

Evaluate and invert.

(B1)

$$y = -16x + 16$$

(A1)

(7)

Alternative for $\frac{dx}{dy}$
Use $t = \frac{1}{4}$ to calculate x

$$x = \frac{72}{y^2} - \frac{3}{y}$$

ACF CSO

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(M1)

$$\frac{dx}{dy} = -\frac{144}{y^3} + \frac{3}{y^2}$$

(A1)

$$\left(\frac{dx}{dy} = \frac{3y - 144}{y^3} \right)$$

Correct expression for x from candidate's implicit equation and attempt derivatives

(A1)

$$(ii) \quad y = -16 \times \frac{3}{2} + 16 = -8$$

Substitute $x = \frac{3}{2}$ into candidate's tangent; calculate y

M1

$$\frac{3}{2}(-8)^2 + 3 \times (-8) = 96 - 24 = 72$$

$y = -8$ used to verify 72

A1

2

[11]

Q20.

(a) (i) $y = \frac{4}{3}x - \frac{7}{3}$

$y = \pm \frac{4}{3}x + k$ or $\frac{\Delta y}{\Delta x}$ with 2 **correct** points

M1

$\Rightarrow \text{grad } AB = \frac{4}{3}$

condone slip in rearranging if gradient is correct; condone 1.33 or better

A1

2

(ii) $y = \text{'their grad'} x + c$
and attempt to use $x = 3, y = -5$
or $y - -5 = \text{'their grad } AB' (x - 3)$
or $4x - 3y = k$ and attempt to find k using
 $x = 3$ and $y = -5$

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$$\left. \begin{array}{l} y + 5 = \frac{4}{3}(x - 3) \\ \text{or } y = \frac{4}{3}x - \frac{27}{3} \end{array} \right\}$$

*correct equation in any form but must
simplify -- to +*

A1

$4x - 3y = 27$

*integer coefficients in required form
eg $-8x + 6y = -54$*

A1

3

(b) $4x - 3y = 7$ and $3x - 2y = 4$

$\Rightarrow 8x - 9x = 14 - 12$ etc

*must use **correct pair** of equations and
attempt to eliminate x or y (generous)*

M1

$x = -2$

A1

$y = -5$

or $D(-2, -5)$

A1

3

(c) $4(k - 2) - 3(2k - 3) = 7$

sub $x = k - 2$, $y = 2k - 3$ into $4x - 3y = 7$

$4k - 8 - 6k + 9 = 7$

*and attempt to multiply out with all k
terms on one side (condone one slip)*

M1

$\Rightarrow k = -3$

A1

2

EXAM PAPERS PRACTICE

[10]

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Q21.

(a) $(x - 5)^2 + (y - 8)^2$

B1

$= 25$

condone 5^2

B1

2

(b) (i) $(2 - 5)^2 + (12 - 8)^2$

or $AC^2 = 3^2 + 4^2$

$= 9 + 16 = 25$

hence $AC = 5$; (also radius = 5)

$\Rightarrow A$ lies on circle

CSO

$(\Rightarrow \text{radius} = AC) \Rightarrow A \text{ lies on circle}$

B1

(must have concluding statement and circle equation correct if using equation)

(must have concluding statement & RHS of circle equation correct or $r = 5$ stated if Pythagoras is used)

1

(ii) $\text{grad} = \frac{4}{3}$

B1

Gradient of tangent is $\frac{3}{4}$

FT their $-1/\text{grad } AC$

B1 ✓

$y - 12 = \text{'their tangent grad'} (x - 2)$

or $y = \text{'their tangent grad'} x + c$

& attempt to find c using $x = 2, y = 12$

M1

$y - 12 = \frac{3}{4}(x - 2) \quad \text{or} \quad y = \frac{3}{4}x + \frac{21}{2} \quad \text{etc}$

correct equation in any form

A1

$3x - 4y + 42 = 0$

CSO must have integer coefficients with all terms on one side of equation accept $0 = 8y - 6x - 84$ etc

A1

5

(c) (i) $(CM^2 =) (7 - 5)^2 + (12 - 8)^2$

or $(CM^2 =) 20$

M1

$(\Rightarrow CM = \sqrt{20}) \Rightarrow (CM =) 2\sqrt{5}$

A1

2

(ii) $PM^2 = PC^2 - CM^2 = 25 - 20$

Pythagoras used correctly eg $d^2 + (2\sqrt{5})^2 = 5^2$

M1

$\Rightarrow PM = \sqrt{5}$

A1

Area $\Delta PCQ = \sqrt{5} \times 2\sqrt{5} = 10$
CSO

A1

3

[13]

Q22.

(a) (i)

$$\left. \begin{aligned} &(\text{Increasing} \Rightarrow) \frac{dy}{dx} > 0 \\ &20x - 6x^2 - 16 > 0 \end{aligned} \right\} \text{ either}$$
correct interpretation of y increasing

M1

$\Rightarrow 6x^2 - 20x + 16 < 0 \quad \text{or} \quad (2) \quad (10x - 3x^2 - 8) > 0$
must see at least one of these steps before final answer for A1

$\Rightarrow 3x^2 - 10x + 8 < 0$
 CSO **AG** no errors in working

A1

2

(ii) $(3x - 4)(x - 2)$

correct factors or correct use of quadratic equation

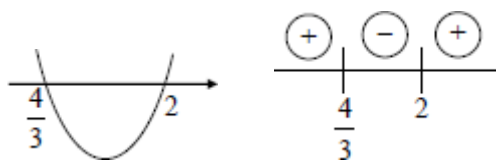
formula as far as $\frac{10 \pm \sqrt{4}}{6}$

M1

CVs are $\frac{4}{3}$ and 2

condone $\frac{8}{6}$ and $\frac{12}{6}$ *here but not in final line*

A1



sketch or sign diagram

M1

$$\frac{4}{3} < x < 2$$

$$\text{or } 2 > x > \frac{4}{3}$$

$$\text{accept } x < 2 \text{ AND } x > \frac{4}{3}$$

$$\text{but not } x < 2 \text{ OR } x > \frac{4}{3}$$

$$\text{nor } x < 2, x > \frac{4}{3}$$

A1

Mark their final line as their answer

4

(b) (i) $x = 2$; $\left(\frac{dy}{dx} =\right) 40 - 24 - 16$

sub $x = 2$ into $\frac{dy}{dx}$ and simplify terms

M1

$$\frac{dy}{dx} = 0 \Rightarrow \text{tangent at P is parallel to the } x\text{-axis}$$

must be all correct working plus statement

A1

2

(ii) $x = 3$; $\frac{dy}{dx} = 20 \times 3 - 6 \times 3^2 - 16$

must attempt to sub $x = 3$ into $\frac{dy}{dx}$

M1

$$(= 60 - 54 - 16) = -10$$

A1

$$\text{Gradient of normal} = \frac{1}{10}$$

$$\frac{-1}{\text{"their"} - 10}$$

A1 ✓

Normal: $(y - -1) = \text{'theirgrad'}(x - 3)$

normal attempted with correct coordinates

used and gradient obtained from their $\frac{dy}{dx}$ value

m1

$$y + 1 = \frac{1}{10}(x - 3)$$

any correct form, eg $10y = x - 13$ but must simplify -- to +

A1

(Equation of tangent at P is) $y = 3$

B1

$$x = 43$$

$$\text{CSO; } \Rightarrow R(43, 3)$$

A1

7

[15]

Q23.

(a) (i) $y = \frac{1}{2}(7 - 3x)$

attempt at $y = \dots$

or use of 2 correct points using $\frac{\Delta y}{\Delta x}$

M1

$$\Rightarrow \text{gradient} = -\frac{3}{2}$$

condone slip in rearranging if gradient is correct

A1

2

(ii) $y = \text{'their grad'} x + c$
and substitution of $x = 2, y = -7$

or using $3x + 2y = k$ with $x = 2, y = -7$
and attempt to find k or $y - -7 = \text{'their grad' } (x - 2)$

M1

$$y = -\frac{3}{2}x + c, c = -4$$

correct equation in any form

$$y + 7 = -\frac{3}{2}(x - 2), 3x + 2y + 8 = 0, \text{ etc}$$

A1

$$(x = 0 \Rightarrow) y = -4$$

or y -intercept = -4 or $D(0, -4)$

A1cso

3

(b) $3x + 2(1 - 4x) = 7, y = 1 - \frac{4}{3}(7 - 2y)$
elimination of y (or x) (condone one slip)

M1

$$x = -1$$

one coordinate correct

A1

$$y = 5$$

other coordinate correct

coordinates of $A(-1, 5)$

A1

3

(c) $(5 - 2)^2 + (k + 7)^2 = 5^2$

condone one sign slip within one bracket

M1

(or $k + 7 = 4$ or $k + 7 = -4$)

$$k = -3$$

one correct value of k

A1

or $k = -11$

both correct (and no other values)

A1

3

[11]

Q24.

(a) (i) $\frac{dy}{dx} = -1 - 4x^3$

one of these terms correct

M1

all correct (no + c)

A1

(When $x = 1$, grad =) -5

(Check that $\frac{dy}{dx}$ is actually correct!)

A1cso

3

(ii) $y - 12 = \text{'their grad'} (x - 1)$

*any form of equation through (1, 12) and
attempt at c if using $y = mx + c$*

M1

$y = -5x + 17$ (or $y = 17 - 5x$)

FT their gradient

Condone $y = -5x + c$, $c = 17$ etc

A1 ✓

2

(b) (i) $14x - \frac{x^2}{2} - \frac{x^5}{5}$

one of these terms correct

M1

another term correct

A1

all correct (may have + c)

A1

$$[I]_2 =$$

$$\left(14 - \frac{1}{2} - \frac{1}{5}\right) - \left(-28 - 2 + \frac{32}{5}\right)$$

F(1) and F(-2) attempted

m1

$$= 36.9 \quad \text{OE}$$

Condone recovery to this value

A1

5

(ii) $\text{Area } \Delta = \frac{1}{2} \times 3 \times 12$

Correct area of triangle unsimplified

M1

$$= 18$$

$$\Rightarrow \text{shaded area} = 18.9$$

A1cso

2

[12]

Q25.

(a) (i) $(x+3)^2 + (y-1)^2$
condone $(x-3)^2$

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$$= 13$$

$$\text{condone } (\sqrt{13})^2$$

B1

2

(ii) $x^2 + 6x + 9 + y^2 - 2y + 1$

*attempt to multiply out both of 'their' brackets;
 must have x and y terms*

M1

$$x^2 + y^2 + 6x - 2y$$

both $m = 6$ and $n = -2$

A1

$$-3 = 0$$

All correct, $p = -3$ and ... = 0

A1

3

(b) $x = 0 \Rightarrow y^2 - 2y - 3 = 0 \Rightarrow (y - 3)(y + 1) = 0$

*putting $x = 0$ PI
and attempt to solve or factorise*

M1

$$y = 3, y = -1$$

A1

$$\Rightarrow \text{Distance } AB = 3 + 1 = 4$$

OR Pythagoras $d^2 = 13 - 3^2$ M1

$$d = 2$$
 A1

$$\text{distance} = 2 \times 2 = 4$$
 A1

A1cso

3

(c) (i) $(-5 + 3)^2 + (-2 - 1)^2 = 4 + 9 = 13$

Substitution $x = -5, y = -2$ into any correct circle equation

$\Rightarrow D$ lies on circle

*convincing verification **plus statement***

B1

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1

(ii) $\text{grad } CD = \frac{1+2}{-3+5}$

condone one sign slip

M1

$$= \frac{3}{2} \quad (\text{or } 1.5)$$

$$\text{not } \frac{-3}{-2}$$

A1

2

- (iii) Perpendicular gradient $= -\frac{2}{3}$
ft their grad CD or $m_1 m_2 = -1$ stated

M1

Tangent has equation $y + 2 = -\frac{2}{3}(x + 5)$

any form of correct equation eg $2x + 3y + 16 = 0$

$$y = -\frac{2}{3}x + c, \quad c = -\frac{16}{3}$$

A1

2

[13]

Q26.

(a) $y = \frac{13}{3} - \frac{7}{3}x$

attempt at $y = a + bx$ or $\frac{\Delta y}{\Delta x}$ with 2 correct points

M1

(gradient =) $-\frac{7}{3}$

condone slip in rearranging if gradient is correct

A1

2

- (b) (i) $y - 3 = \text{'their grad'}(x - -1)$
or $7x + 3y = k$ and attempt at k using
 $x = -1$ and $y = 3$
or $y = (\text{their } m)x + c$ and attempt at c
using $x = -1$ and $y = 3$

M1

$$y - 3 = -\frac{7}{3}(x + 1) \text{ or } 7x + 3y = 2 \text{ or } y = -\frac{7}{3}x + c, \quad c = \frac{2}{3}$$

correct equation in any form and replacing
-- with + sign

A1cso

2

(ii) $(4, -5)$

$x = 4$, $y = -5$

withhold if clearly from incorrect working

B1,B1

2

(c) $7x + 3y = 13$ and $3x + 2y = 12$

$\Rightarrow$ equation in x or y only

*must use correct pair of equations and
attempt to eliminate y (or x)*

M1

$x = -2$

A1

$y = 9$

A1

3

[9]

Q27.

(a) $(x - 3)^2 + (y + 8)^2$

accept $(y - -8)^2$

B1

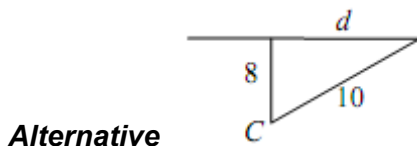
$= 100$

condone $RHS = 10^2$ or $k = 10^2$

B1

2

(b) $y = 0 \Rightarrow$ 'their' $(x - a)^2 + b^2 = k$



M1

$(x - 3)^2 = 36$ or $x^2 - 6x - 27 (= 0)$ (PI)

A1

$$\Rightarrow x = -3, 9$$

$$(d^2 =) 10^2 - 8^2 \quad M1$$

$$d^2 = 36 \quad A1 \quad \text{or } d = 6$$

$$\Rightarrow x = -3, 9 \quad A1$$

A1

3

(c) Line CA has gradient $-\frac{2}{5}$

M1

$$CA \text{ has equation } (y+8) = -\frac{2}{5}(x-3)$$

any form of correct equation

$$\text{eg } y = -\frac{2}{5}x + c, \quad c = -\frac{34}{5}$$

A1

$$2x + 5y + 34 = 0$$

integer coefficients – all terms on 1 side

A1cso

3

(d) (i) their $(x-3)^2 + (2x+1+8)^2$ or

$$x^2 + (2x+1)^2 - 6x + 16(2x+1) + 73$$

*substituting $y = 2x + 1$ correctly into LHS
of “their” circle equation and attempt to
expand in terms of x only*

M1

$$x^2 - 6x + 9 + 4x^2 + 36x + 81 = 100$$

$$\text{or } x^2 + 4x^2 + 4x + 1 - 6x + 32x + 16 + 73 = 100$$

any correct equation (with brackets expanded)

A1

$$\Rightarrow 5x^2 + 30x - 10 = 0$$

must see this line or equivalent

$$\Rightarrow x^2 + 6x - 2 = 0$$

AG; all algebra must be correct

A1cso

3

(ii) $(x + 3)^2 = 11$

or correct use of formula

must get as far as $x = \frac{-6 \pm \sqrt{44}}{2}$

M1

$x = -3 \pm \sqrt{11}$

exactly this

A1cso

2

[13]

Q28.

(a)

For either 6 or $6x^0$

B1

$\frac{dy}{dx} = 6 - 3x^{\frac{1}{2}}$

$Ax^{\frac{3}{2}-1}$, $A \neq 0$ OE

M1

$6 - 3x^{\frac{1}{2}}$ or $6 - 3\sqrt{x}$ with no '+c'

[If unsimplified here, A1 can be awarded retrospectively if correct simplified expression is seen explicitly in (b)(i).]

A1

3

(b) (i) $6 - 3x^{\frac{1}{2}} = 0$

Equating c's $\frac{dy}{dx}$ to 0 PI by correct ft rearrangement of c's $\frac{dy}{dx} = 0$

M1

$$x^{\frac{1}{2}} = 2 \Rightarrow x = 2^2$$

$$x^{\frac{1}{2}} = k \quad (k > 0), \text{ to } x = k^2.$$

PI by correct value of x if no error seen

m1

$M(4, 8)$

SC If M0 award B1 for (4, 8)

A1

3

(ii) Eqn of normal at M is $x = 4$

Ft on $x = c$'s x_M

B1F

1

(c) (i) When $x = \frac{9}{4}, \frac{dy}{dx} = 6 - 3 \times \frac{3}{2} = \frac{3}{2}$

Attempt to find $\frac{dy}{dx}$ when $x = \frac{9}{4}$

M1

Gradient of normal at $P = -\frac{2}{3}$

$m \times m' = -1$ used

m1

Eqn of normal: $y - \frac{27}{4} = -\frac{2}{3} \left(x - \frac{9}{4} \right)$

ACF eg $y = -\frac{2}{3}x + \frac{33}{4}$

A1

$$12y - 81 = -8x + 18 \Rightarrow 8x + 12y = 99$$

Coeffs and constant must now be positive integers, but accept different order

eg $12y + 8x = 99$

A1

4

(ii) $8(4) + 12y = 99$

*Solving c's answer (b)(ii), (must be in form $x = \text{positive const}$), with c's answer (c)(i).
 PI by correct earlier work and correct coordinates for R.*

M1

$$R\left(4, \frac{67}{12}\right)$$

Accept 5.58 or better as equivalent to $\frac{67}{12}$

A1

2

[13]

Q29.

- (a) Use of one law of logs or exponentials

M1

$$\lg a = c \text{ and } \lg b = m$$

OE; both needed

A1

$$\text{So } a = 10^c \text{ and } b = 10^m$$

A1

3

- (b) Points (1, 1.08), (5, 1.43) plotted

M1 A0 if one point correct

M1A1

Straight line drawn through points

ft small inaccuracy

A1F

3

- (c) (i) Attempt at antilog of $Y(3)$

OE

M1

$$\text{When } x = 3, Y \approx 1.25 \text{ so } y \approx 18$$

Allow AWRT 18

A1
2

- (ii) Attempt at a as antilog of Y -intercept
OE

M1

$$a \approx 9.3 \text{ to } 10$$

AWRT

A1
2

[10]

Q30.

(a) (i) $\text{grad } AB = \frac{7-3}{3-1}$

$\frac{\Delta y}{\Delta x}$ correct expression, possibly implied

M1

= 2 (must simplify 4/2)

A1
2

(ii) $\text{grad } BC = \frac{7-9}{3+1} = -\frac{2}{4}$

Condone one slip

NOT Pythagoras or cosine rule etc

M1

$$\text{grad } AB \times \text{grad } BC = -1$$

$$\Rightarrow \angle ABC = 90^\circ \text{ or } AB \text{ \& } BC \text{ perpendicular}$$

convincingly proved plus statement

SC B1 for $-1/(\text{their grad } AB)$

or statement that $m_1 m_2 = -1$ for perpendicular lines if M0 scored

A1
2

- (b) (i) M 0,6
- B1 + B1 each coordinate correct

B2
2

(ii)
$$\left. \begin{aligned} AB^2 &= 3 - 1^2 + 7 - 3^2 \\ BC^2 &= 3 + 1^2 + 7 - 9^2 \end{aligned} \right\}$$

either expression correct, simplified or unsimplified

M1

$AB^2 = 2^2 + 4^2$ or $BC^2 = 4^2 + 2^2$

or $\sqrt{20}$ found as a length

Must see either $AB^2 = \dots$, or $BC^2 = \dots$,

A1

$$\left. \begin{aligned} AB^2 &= BC^2 \Rightarrow AB = BC \\ \text{or } AB &= \sqrt{20} \text{ and } BC = \sqrt{20} \end{aligned} \right\}$$

A1

3

(iii)
$$\text{grad } BM = \frac{7-6}{3-0}$$

ft their M coordinates

M1

or $-1/(\text{grad } AC)$ attempted $= \frac{1}{3}$

correct gradient of line of symmetry

A1

BM has equation $y = \frac{1}{3}x + 6$

CSO, any correct form

A1

3

[12]

Q31.

(a) $x^2 - 8x + 15 + 2$

Terms in x must be collected, PI

B1

their $x - 4^2$ (+ k)

ft $x - p^2$ for their quadratic

M1

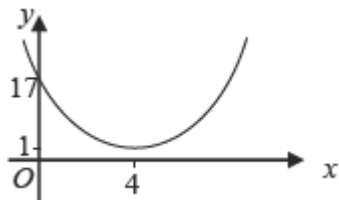
$$= x - 4^2 + 1$$

*ISW for stating if $p = -4$ correct
expression seen*

A1

3

(b) (i)



U shape in any quadrant (generous)

M1

*correct with min at (4, 1) stated or 4 and 1
marked on axes
condone within first quadrant only*

A1

*crosses y-axis at (0, 17) stated or 17
marked on y-axis*

B1

3

(ii) $y = k$

$y = \text{constant}$

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M1

$$y = 1$$

Condone $y = 0x + 1$

A1

2

(c) Translation *(not shift, move etc)*
and no other transformation

E1

with vector $\begin{bmatrix} 4 \\ 1 \end{bmatrix}$

*One component correct
or ft either their p or q*

M1

CSO; condone 4 across, 1 up; or two separate vectors etc

A1

3

[11]

Q32.

(a) (i) $\frac{dy}{dx} = 24x - 19 - 6x^2$

2 terms correct

M1

all correct (no + c etc)

A1

when $x = 2, \frac{dy}{dx} = 48 - 19 - 24$

m1

$\Rightarrow$ gradient = 5
CSO

A1

4

(ii) grad of normal = $-\frac{1}{5}$
ft their answer from (a)(i)

B1ft

$$y + 6 = \left(\text{their} - \frac{1}{5} \right) x - 2$$

or $y = \left(\text{their} - \frac{1}{5} \right) x + c$ and c evaluated

ft grad of their normal using **correct** coordinates BUT must not be tangent
condone omission of brackets

M1

using $x = 2$ and $y = -6$

$$x + 5y + 28 = 0$$

CSO; condone all on one side in different order

A1

3

(b) (i) $\frac{12}{3}x^3 - \frac{19}{2}x^2 - \frac{2}{4}x^4$

one term correct

M1

another term correct

A1

all correct (ignore + c or limits)

A1

$= 32 - 38 - 8$

F 2 attempted

m1

$= -14$

CSO; withhold A1 if changed to +14 here

A1

5

(ii) Area $\Delta = \frac{1}{2} \times 2 \times 6 = 6$
condone -6

B1

Shaded region area = $14 - 6$

difference of $\pm \int \pm \Delta$

$= 8$

CSO

A1

3

[15]

Q33.

(a) $y = \frac{14}{3} - \frac{2}{3}x$

Attempt at y = ...

M1

Gradient $AB = -\frac{2}{3}$

Condone error in rearranging equation

A1

2

- (b) (i) $y - 7 = \text{"their grad } AB" (x - 3)$
or $2x + 3y = k$ and sub $x = 3, y = 7$
or $y = mx + c, m = \text{their grad } AB$ and attempt to find c using $x = 3, y = 7$

M1

$$y - 7 = -\frac{2}{3}(x - 3) \quad \text{OE}$$

$$2x + 3y = 27, \quad y = -\frac{2}{3}x + 9 \text{ etc}$$

A1

2

- (ii) $m_1 m_2 = -1$
or negative reciprocal (stated or used PI)

M1

$$\Rightarrow \text{grad } AD = \frac{3}{2}$$

FT their grad AB

A1ft

$$y - 7 = \frac{3}{2}(x - 3)$$

Any correct equation unsimplified

A1

$$\Rightarrow 3x - 2y + 5 = 0$$

Integer coefficients; all terms on one side, condone different order or multiples.

$$\text{eg } 0 = 4y - 6x - 10$$

A1

4

- (c) $2x + 3y = 14$ and $5y - x = 6$ used
 with x or y eliminated (generous)

$$2(5y - 6) + 3y = 14 \quad \text{etc}$$

M1

$$x = 4,$$

A1

$$y = 2$$

B(4, 2) full marks NMS

A1

3

[11]

Q34.

(a) $(x + 5)^2 + (y - 6)^2 = 5^2$

One term correct LHS

M1

LHS all correct

A1

RHS correct: condone = 25

B1

3

(b) (i) sub $x = -2, y = 2$ into circle equation

Circle equation must be correct

$$3^2 + (-4)^2 = 25 \Rightarrow \text{lies on circle}$$

Must have concluding statement

B1

1

(ii) Grad $PC = -\frac{4}{3}$

Condone $\frac{4}{-3}$

B1

Normal to circle has equation

$$y - 6 = \text{'their gradient } PC' (x + 5)$$

$$\text{or } y - 2 = \text{'their gradient } PC' (x + 2)$$

M0 if tangent attempted or incorrect coordinates used

M1

$$y - 6 = -\frac{4}{3}(x + 5)$$

$$\text{or } y - 2 = -\frac{4}{3}(x + 2)$$

Any correct form eg $4x + 3y + 2 = 0$

$$y = -\frac{4}{3}x + c, \quad c = -\frac{2}{3}$$

A1cso

3

$$(iii) \quad PM = \frac{1}{2} \times \text{radius}$$

M1

$$= 2.5$$

A1cso

$$PO = \sqrt{8}$$

B1

P is closer to the point M

E1 cso

Alternative 1

Attempt at $M\left(-\frac{7}{2}, 4\right)$ with at least one correct coordinate **and** PM^2 attempted

(M1)

$$PM^2 = \frac{9}{4} + 4 = \frac{25}{4}$$

(A!cso)

$$PO^2 = 4 + 4 = 8$$

(B1)

Statement following correct values

(E1cso)

Alternative 2

Attempt at $M\left(-\frac{7}{2}, 4\right)$ with at least one correct coordinate **and** attempt at vectors or difference of coordinates

(M1)

$$\overrightarrow{PM} = \begin{pmatrix} -1.5 \\ 2 \end{pmatrix} \quad OE$$

(A1cso)

P is closer to the point M

(E1cso)

Components of their $\overrightarrow{PM}$ and $\vec{OP}$ considered – **totally independent** of M1

(E1)

4

[11]

Q35.

(a) $M(3, 2)$

B1 for each coordinate

B1B1

2

(b) Gradient $AB = \frac{-2-6}{5-1} = \left(\frac{-8}{4}\right)$

May use coords of M instead of A or B - condone one slip

M1

$= -2$

CSO Answer must be simplified to -2

A1

2

(c) (i) Gradient of perpendicular = $\frac{1}{2}$
ft "their" – $1/\text{gradient } AB$

B1ft

$$\Rightarrow y - 2 = \frac{1}{2}(x - 3)$$

attempt at perp to AB; ft their M coords

M1

$$\Rightarrow 2y - 4 = x - 3 \Rightarrow x - 2y + 1 = 0 \quad \mathbf{AG}$$

CSO Must write down the printed answer

A1

3

(ii) $k - 2(k + 5) + 1 = 0$ or $\frac{(k + 5) - 2}{k - 3} = \frac{1}{2}$

Sub into given line equation or correct expression involving gradients

Condone omission of brackets or use of x

M1

$$\Rightarrow k = -9$$

Condone $x = -9$

(Full marks for correct answer without working)

A1

2

[9]